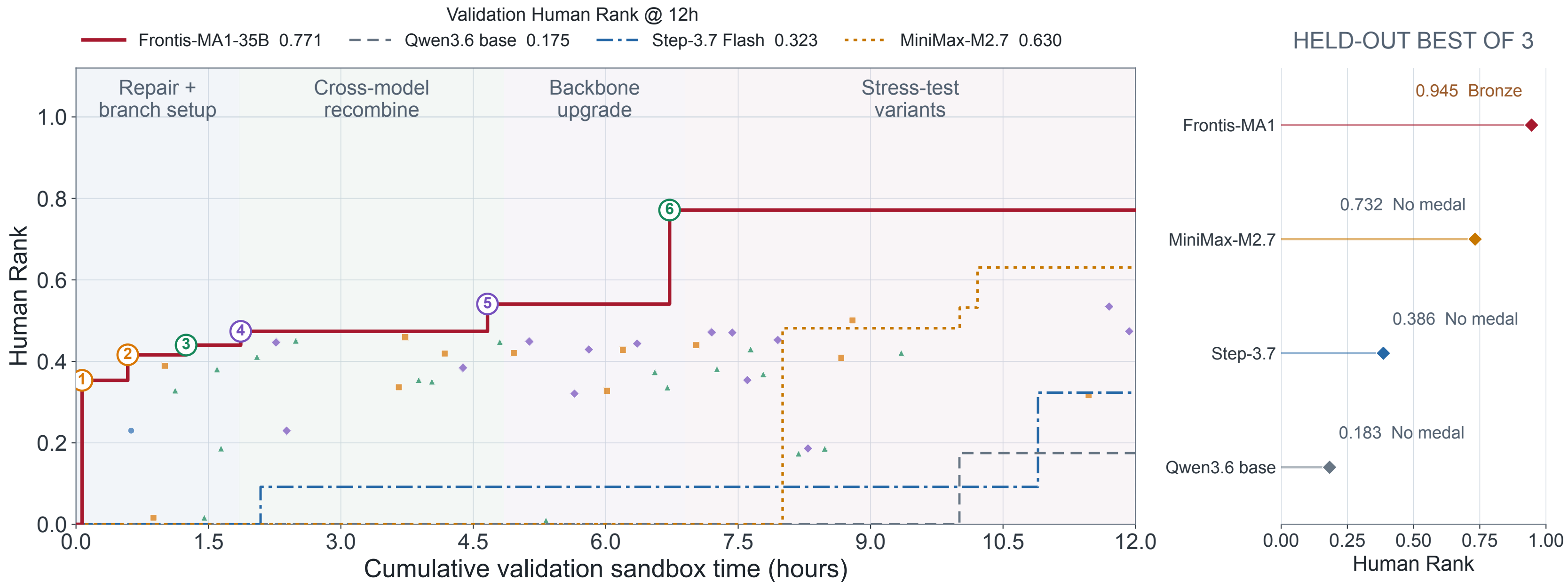


Late Crossover and Improve Transform a Leaf Classifier

leaf-classification • epoch 0 • OpenMLE-Evo • Frontis-MA1 nodes with matched baselines



Frontis-MA1 Milestones

Key program changes at the numbered trajectory nodes.

● Draft ■ Debug ▲ Improve ◆ Crossover

1 DEBUG: VALID CV

Fixes multiclass label encoding, unstable folds, and LightGBM regularization.

step 2 Log loss 0.44622 Human Rank 0.3534

2 DEBUG: STABLE IMAGE

Covers all 99 classes and removes irrelevant color jitter from the ResNet18 branch.

step 4 Log loss 0.23730 Human Rank 0.4160

3 IMPROVE: MULTIMODAL

Combines EfficientNet embeddings with 192 engineered margin, shape, and texture features.

step 11 Log loss 0.17472 Human Rank 0.4398

4 CROSSOVER: HYBRID FUSION

Fuses the robust ResNet18 branch with regularized LightGBM features in one hybrid model.

step 15 Log loss 0.13123 Human Rank 0.4737

5 CROSSOVER: ROBUST FUSION

Combines multimodal fusion, stronger augmentation, early stopping, and TTA.

step 29 Log loss 0.08268 Human Rank 0.5407

6 IMPROVE: CONVNEXT LEAP

Uses ConvNeXt-Tiny embeddings with a regularized MLP over fused image and tabular features.

step 45 Log loss 0.02990 Human Rank 0.7713